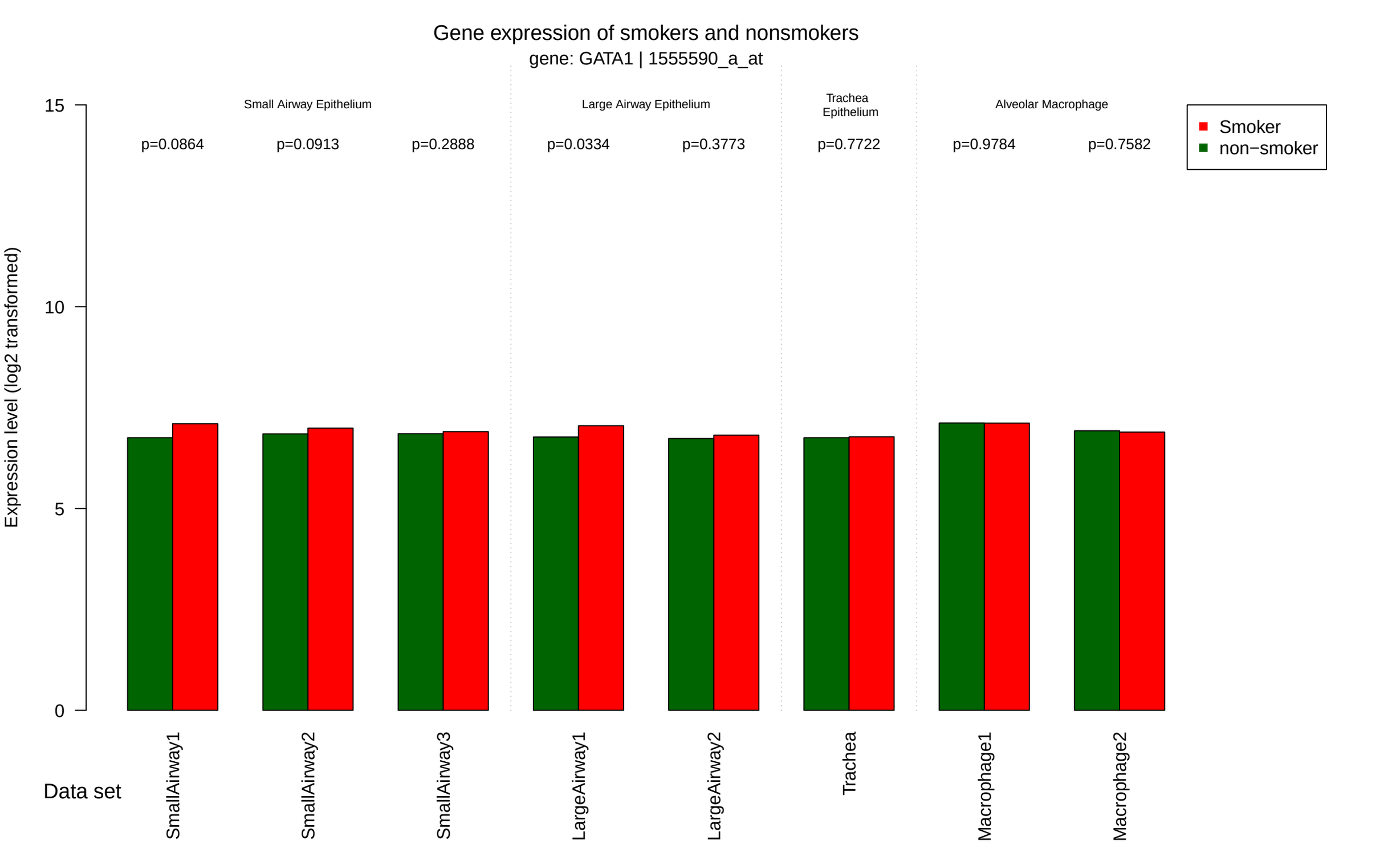
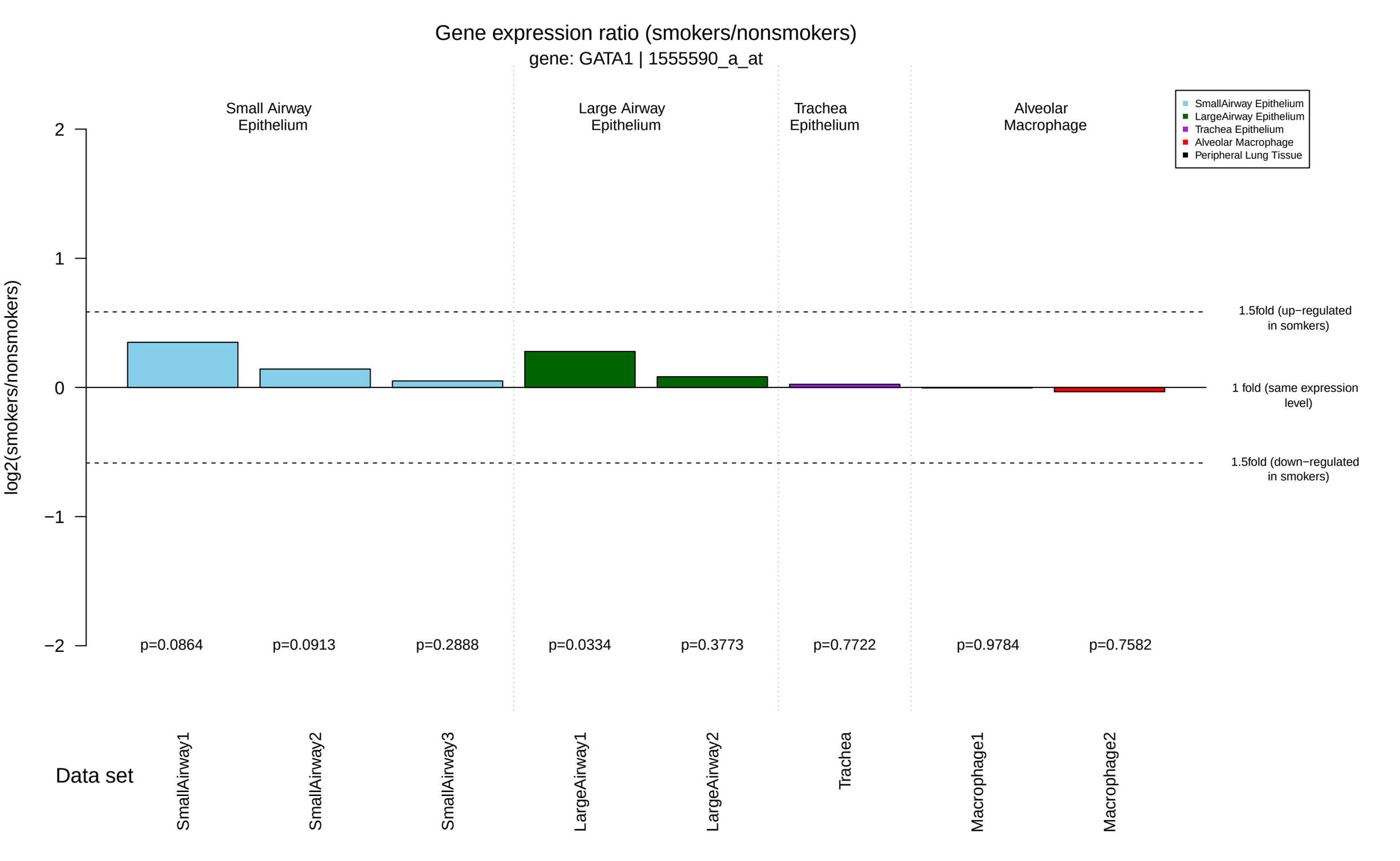
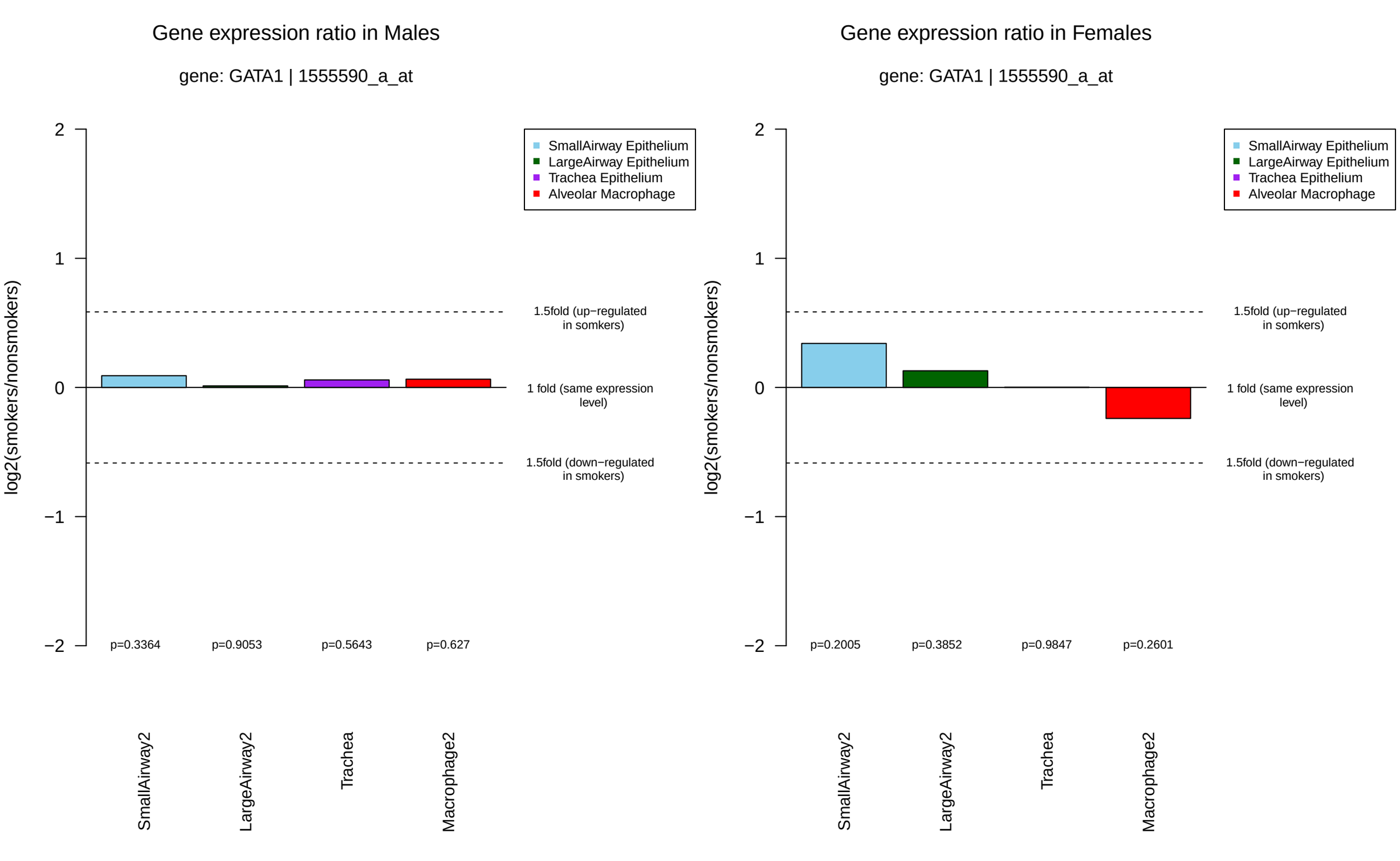
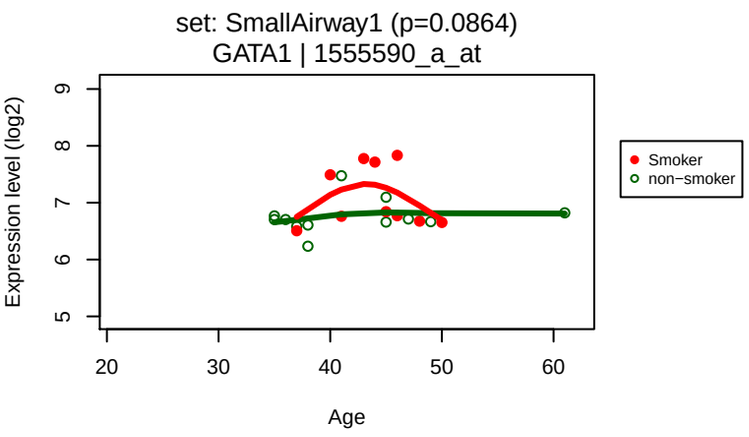
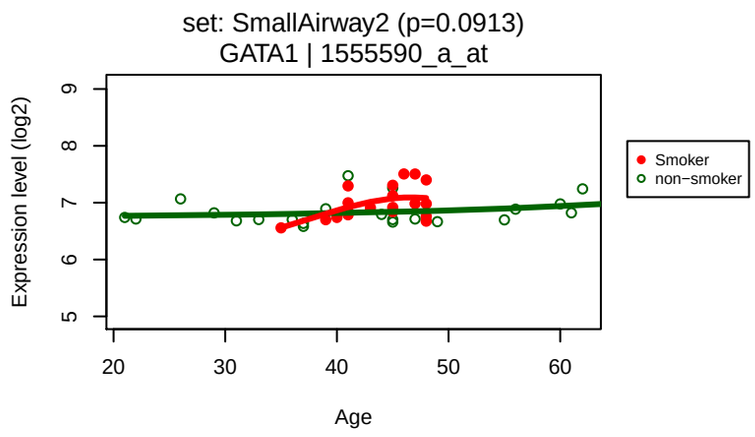
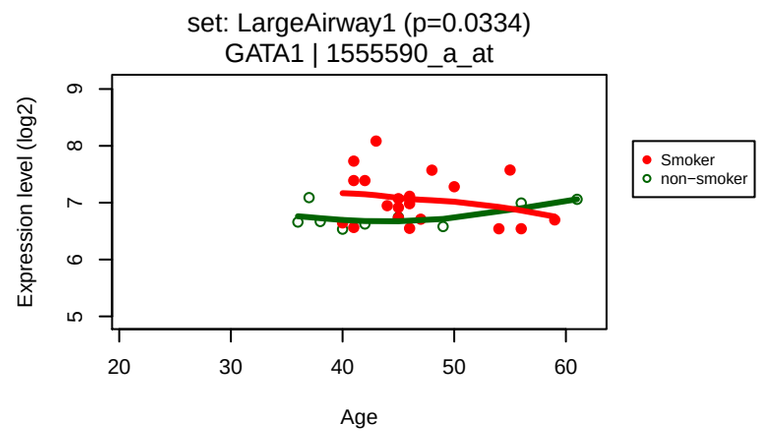

Gene expression of smokers and nonsmokers
gene: GATA1 | 1555590_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0864
p=0.0913
p=0.2888
p=0.0334
p=0.3773
p=0.7722
p=0.9784
p=0.7582
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

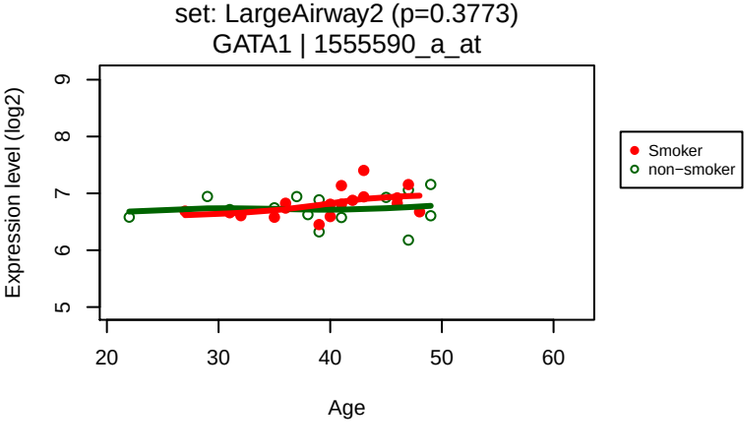

Gene expression ratio (smokers/nonsmokers)
gene: GATA1 | 1555590_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0864
p=0.0913
p=0.2888
p=0.0334
p=0.3773
p=0.7722
p=0.9784
p=0.7582
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

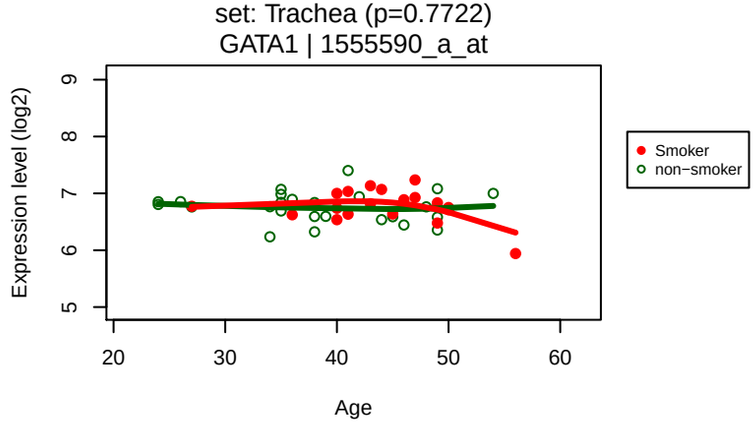

Gene expression ratio in Males
gene: GATA1 | 1555590_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3364
p=0.9053
p=0.5643
p=0.627
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATA1 | 1555590_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2005
p=0.3852
p=0.9847
p=0.2601
SmallAirway2
LargeAirway2
Trachea
Macrophage2

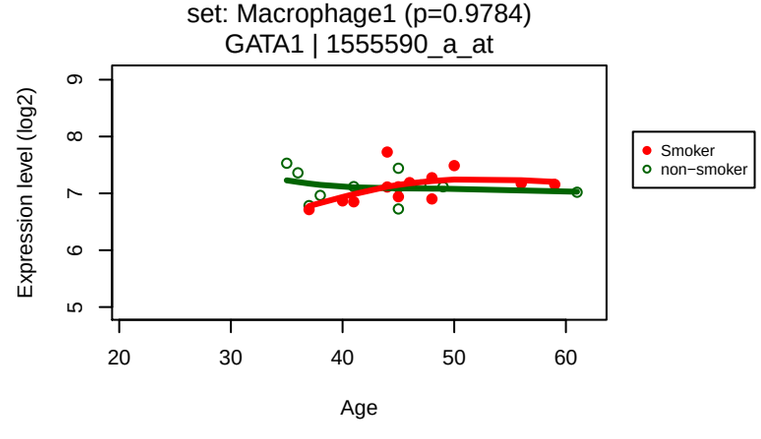

set: SmallAirway1 (p=0.0864)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker

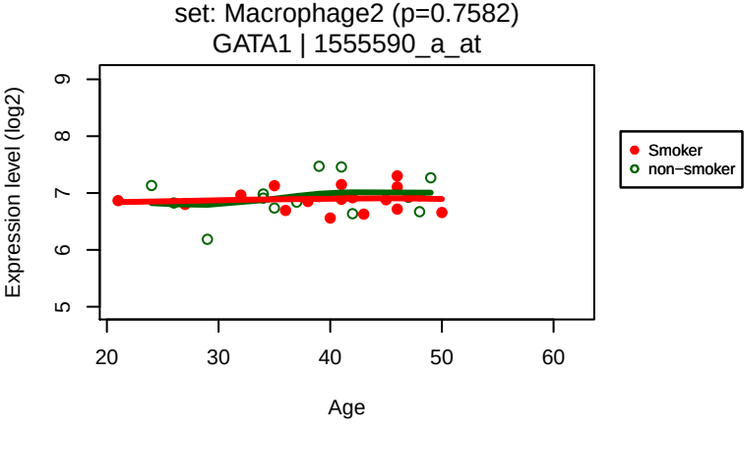

set: SmallAirway2 (p=0.0913)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker

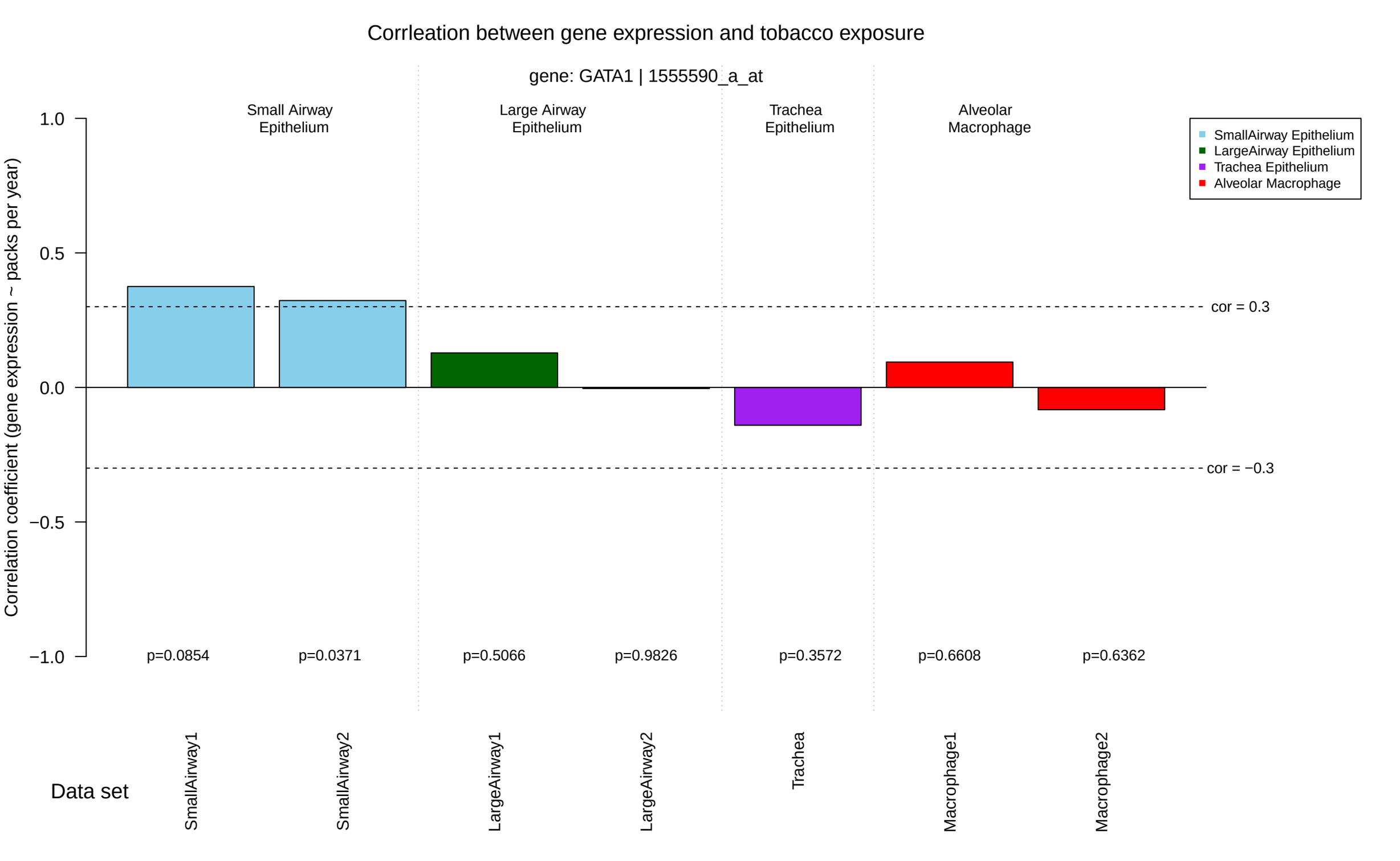

set: LargeAirway1 (p=0.0334)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3773)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7722)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9784)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7582)
GATA1 | 1555590_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATA1 | 1555590_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0854
p=0.0371
p=0.5066
p=0.9826
p=0.3572
p=0.6608
p=0.6362
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2